

Mohit Kumar

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PROFESSIONAL SUMMARY

Full Stack Developer with hands-on experience building responsive, high-performance web applications using React.js, Node.js, JavaScript, and MongoDB. Strong understanding of component-based architecture, backend systems, and real-time data flow. Experienced in designing scalable systems, handling continuous data streams, and optimizing performance. Proven ability to build end-to-end applications with focus on system architecture, efficiency, and real-world problem solving.

EDUCATION

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|---|------------------|
| Jaypee University of Information Technology | Himachal Pradesh |
| • <i>B.Tech in Computer Science and Engineering (GPA: 7.67/10)</i> | 2022 – 2026 |
| Coursework: Data Structures and Algorithms, Artificial Intelligence, Cloud Computing, Computer Networks | |
| Kairali School | Ranchi |
| • <i>Class 12 (CBSE) – 81.5%</i> | 2022 |
| Kairali School | Ranchi |
| • <i>Class 10 (CBSE) – 92.4%</i> | 2020 |

SKILLS SUMMARY

- **Languages:** C++, C, Python, JavaScript, SQL
- **Full Stack:** React.js, Node.js, MongoDB, REST APIs, HTML5, CSS3, Tailwind CSS
- **Systems:** System Architecture (Basic), Real-time Systems, WebSockets

PROJECTS

- **Real-Time Bus Tracking System:** Built a full-stack real-time tracking system using React, Node.js, and WebSockets to simulate continuous vehicle movement. Designed scalable client-server architecture to handle real-time data streams with low latency. Implemented event-driven communication and optimized update frequency to reduce network overhead and improve responsiveness. Demonstrates strong understanding of system architecture, real-time data handling, and scalability.
Tech: Node.js, Socket.io, React.js
- **Lane-Aware Trajectory Prediction for Autonomous Vehicles:** Developed an end-to-end lane-aware computer vision pipeline for autonomous driving by processing recorded road video streams in real time. Applied image preprocessing and perspective transformation techniques to improve lane boundary extraction accuracy under varying lighting and road conditions. Computed lane curvature and vehicle lateral offset to support steering and trajectory alignment, reducing lane deviation by approximately 25%. Achieved consistent real-time inference at 20 FPS on high-resolution video frames while validating results against sample scenarios.
Tech: Python, OpenCV, NumPy, nuScenes
- **Portfolio Website:** Designed and developed a personal portfolio using modular React components. Focused on performance optimization, scalability, and clean architecture for reusable UI components.
Tech: React.js, JavaScript

CERTIFICATIONS

- : Machine Learning Foundation Certification – Infosys Springboard
- : Explore Machine Learning Using Python – Infosys Springboard

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

- **Frontend Developer – IEEE JUIT Student Branch (2023–2024):** Developed and maintained frontend components for technical platforms and collaborated with teams to deliver scalable and responsive UI solutions.

CORE COMPETENCIES

- Strong problem-solving skills using Data Structures and Algorithms
- Full-stack development with focus on scalable systems
- Real-time system design and performance optimization
- Understanding of system architecture and backend communication
- Ability to learn quickly and work on production-level systems